Stochastic modeling for describing crystallization droplets in water emulsion

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Abstract

This paper introduces a novel, stochastic mathematical model for the crystallization of emulsion in a dispersed media. The new mathematical model reads as a stochastic partial differential equations by combining the heat energy equation and the nucleation theory with the specified drift and diffuion. We show the existence and uniqueness of the solution to mathematical model by using some techniques of stochastic partial differential equations. Numerical experiments are drawn to support the theoretical results and to compare our numerical results to experimental one.

Keywords: mathematical model, stochastic differential equation, Brownian motion, exitence and uniqueness, crystallization, emulsion.

1. Introduction

1.1. Background

In mathematical modeling a Brownian dynamical problem, a stochastic differential equations is often developed, analyzed and studied to describe different phenomena. The stochastic partial differential equations, in addition, are very studied recently and there are many open problems in this research topic. There appear to be three ways for using and developing stochastic differential equation models for many applications in engineering, chemistry, biology, geology, financial mathematics and physics. In the first modelling method, many authors [23, 44–46] developed a discrete stochastic model by studying changes in the system components over a small time ranges. This method is a natural one used for many years in modelling deterministic dynamical systems in engineering and physics, where changes in the model are treated over a very small time intervall and a differential equation is found when the time interval

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that approaches to zero. Similarities between the Kolmogorov's Forward equation satisfied by the probability laws of continuous-and discrete-time stochastic models let us conclude that an Itô stochastic model is close to the discrete one. Here, the number of Brownian motion in the stochastic differential equation never exceeds the number of unkown components in the setting system. A second one is devoted to identify all of the different independent random changes that arise in the dynamical system. Suitable terms are identified for these changes in developing a discrete time mathematical model which is approximated by a stochastic differential equations, see for example [47–49]. In this case, the total number of random term changes may exceed the number of unkown components in the dynamical system. This method yields systems of Itô stochastic differential equations (SDE) that are generally easy to discretize and solve numerically. In the third method, the direct formulation of a system of Itô stochastic differential equations is most used in building an Itô SDE. For a given random dynamical system, we assume that the drift and diffusion have the specific functional forms and have frequently a linear form to simplfiy the study, [50–53]. These three methods have been used in modeling many dynamical systems that experience random influences. After the construction of the SDE model, the problem arise in showing that this model admits the unique solution under the initial conditions, we refer to [14, 23, 25, 47, 48], and two essential papers for estimating energy inequality [29, 49]. In our investigation, only the first method is discussed to develop our stochastic model in order to explain the crystallization in water emulsion.

1.2. Earlier work and novel contributions

Water emulsions are systems formed by two immiscible liquids in which one (dispersed phase i.e. water) is dispersed in another (emulsifying support). This dispersion takes place in the form of small water droplets [10, 24]. Water emulsions can be produced by two-stage process emulsification[19, 20]. During the first one, the dispersed phase is emulsified above its fusion temperature and crystallized during cooling in the second one. The analysis of the crystallization of these droplets is of a major importance in various fields of application like food, petroleum industry and pharmacology [2, 30]. During the cooling process, the majority of the droplets remain in a metastable state (i.e. undercooled liquid) and exhibit a stochastic crystallization behaviour [3, 8]. Ideally, after rupture of supercooling, the liquid droplets are transformed into solid particles of the same size. Since the droplets in emulsions are commonly polydisperse and the nucleation in emulsions is strongly droplet size dependent, individual and stochastically distributed crystallization often occurs during cooling in oil-in-water dispersions. Inhomogeneous crystallization leads to the presence of supercooled liquid and crystallized solid droplets within one dispersion and thus to incomplete crystallization [7, 9]. Consequently, supercooled droplets may collide with crystallized particles, wet their surface and crystallize/coalesce with those particles leading to an increase in the particle size distribution [15, 18].

It is interesting to outline that when one liquid is dispersed in into another one, the surface area that separates the two liquids increases considerably, mak-

ing the emulsion unstable [4, 12]. In this case, the coalescence phenomenon becomes a determining factor in the stability of emulsions. This phenomenon will depend on several parameters and especially on temperature. Since droplet crystallization is an exothermic reaction, it can lead to an increase in the temperature of the emulsifying medium and cause a chaotic droplet movement in the emulsion, leading to coalescence. In addition, a high mass fraction of the dispersed liquid increases the contact between the droplets, therefore their coalescence [12].

Any crystallization process starts with nucleation due to the supercooling phenomenon. Usually, nuclei are produced by heterogeneous nucleation mechanism due to the presence of impurities in the liquid droplets [6, 16]. Therefore, it is theoretically proved that when heterogeneous nucleation is considered, crystallization takes place at a higher temperature. In water emulsion, the volume of individual droplets is comparatively low and the number of impurities is usually much smaller than the number of droplets [13]. Consequently, nuclei formation in emulsion may approach the homogenous nucleation mechanism, which requires higher supercooling [4, 12] and can be a major barrier to industrial application [27].

A first investigation of the stochastic perturbation for deterministic model by Brownian motion, in order to describe the fluctuations growth crystallization, is introduced by Vallet and David [29]. The authors proposed the stochastic mathematical model for ratio crystallized doplets φ, but we did not identify exactly the associated term $b(u, \varphi)$ to Brownian motion W_t. The model is written as follows

$$\partial_t \varphi = (1 - \varphi) J(u) + b(u, \varphi) \partial_t W_t \quad \text{in } (0, T] \times \Omega. \tag{1.1}$$

This model is complemented with the heat energy transfer (see a first equation in (2.3)) and the initial and boundary conditions. Vallet and al. have shown the existence and uniqueness of solution under some hypotheses, see [29] for more details. To my best knowledge our this topic, there exists no paper to develop a stochastic model with a specific diffusion term (i.e. term with independent variables). Therefore, the first main of this paper is to carry out the development of a stochastic mathematical model with a specific diffusion term inspired by Black-Scholes model in finance. The second main is the theoretical analysis of existence and uniqueness of solution to our new stochastic model for describing the crystallization of water droplets dispersed in an emulsion.

1.3. Outline of this work

The rest of this paper is structured as follows: We conclude this section by summarizing our notations. In section 2, we construct and develop the stochastic model, and we conclude this section by stating and developing our stochastic model to describe the speed crystallization in dispersed media. Then, in section 4 we present firstly the solution to the stochastic model. In a second time, we explain how to discretize our model, discuss the need for passing a limit in the sequel, and conclude this section by stating the main results of the existence and

uniqueness of the solution are stated by using stochastic differential equation theory. Finally, in section 6 we present a numerical results to confirm the theoretical result, and we draw the conclusion.

1.4. Notations

Throughout this paper, $\Omega \subset \mathbb{R}^d$, $d \geqslant 1$ denotes an open and bounded domain with Lipschitz boundary $\partial\Omega$ and an outward unit normal n. Let $(0,T)$ be a time interval, $0 < T < +\infty$ and let $\Gamma = \partial\Omega$. Moreover, we denotes $(\Theta, \mathcal{F}, \mathbb{P})$ the complete probability space, $\mathbb{E}$ the mathematical expectation (the integral over Ω with respect the probability measure $\mathbb{P}$), $(\mathcal{F}_t)_{0\leqslant t\leqslant T}$ the complete and right continuous filtration contained in $\mathcal{F}$. Furthermore, we will use $(W_t)_{0\leqslant t\leqslant T}$ to denote a standard one-dimensional Brownian motion (Wiener process), and we also assume that it starts at 0. In addition, we consider $\mathbb{H} = L^2(\Theta \times L^2(\Omega))$, $\mathcal{H} = L^2((0,T);\mathbb{H})$, $\mathbb{V} = L^2(\Theta \times H^1(\Omega))$ and $\mathcal{V} = L^2((0,T);\mathbb{V})$. Let $\overline{\lim}$ and $\lim$ be the upper limit and the lower limit when N goes to infinity respectively. We will denote by $\partial_t u$, $\partial_x u$ and $\partial_{xx} u$ the first derivative in time, in space x and the second derivative of u.

Finally, let us denote by $\|\cdot\|_{0,\Omega}$ the norm associated to $L^2(\Omega)$, by $\|\cdot\|_{1,\Omega}$ the norm associated to $H^1(\Omega)$ and by $\|\cdot\|_{\gamma,T,\Omega}$ the norm associated to $\mathcal{H}$ with

$$\|v\|_{\gamma,T,\Omega} = \left(\int_0^T e^{-\gamma t}\mathbb{E}\left(\|v\|_{0,\Omega}^2\right)\, dt\right)^{1/2}. \tag{1.2}$$

Convention: In this work, we will note by C a positive constant and its expression may be dependent of Ω and T and independent of the various indices. Thus, we do not need to precise its expression and it may vary from one line to the other, and even in the same line. The notation $C_{\alpha,\beta,\gamma}$ means that the constant C depends only on α, β and γ.

2. Stochastic modeling of crystallization under some assumptions

2.1. Problem setting: first main result

In emulsion, the crystallization inside each droplet occurs over two different phases: the nucleation process and the crystalline growth, triggered by a supercritical germ, leading to the total solidification of the droplet [8]. In this work, we will mainly focussed on water droplet crystallization dispersed in an emulsion. Due to the relatively small droplet dimensions, we can assume that the crystallization of dispersed water droplets is driven only by the rate of nucleation of crystals within the droplet volume [3]. As you know, there is an enormous work in this topic, see [7–9]; perhaps, the point of special interest in the thesis report by El-Rhafiki [38], and Tembely [28] is the important suggest that the steady cooling of a water emulsion presents the occurrence of supercooling (the persistence of the liquid state of the water below its melting temperature T_m) as well as the unpredictable nature of water crystallization in droplets [28]. This means that water droplets do not simultaneously crystallize

at the same temperature and crystallization inside emulsion takes place over a range of temperatures. Due to the stochastic nature of nucleation, it is not possible to determine the location where crystallization began inside the emulsion. However, a probability of the occurrence of solid droplets can be estimated as it is possible to estimate the nucleation rate [7].

In order to announce our mathematical modelling, we assume that a total of droplets N with volumes V are cooled at a constant cooling rate β. We denote $n_c(t,x)$ a proportion of N which corresponds to the number of crystalized droplets in a control volume at the position x and at the time t. Therefore, the number of crystallized droplets $n_c(t,x)$ is written in differential form, see [38, Section 2.4]:

$$dn_c(t,x) = (N - n_c(t,x))\, J(u)dt \tag{2.1}$$

where u is the temperature which depends on time t and space x, J is the nucleation probability and its form is determined by many authors []. By using the crystallized fraction of droplets $\varphi(t,x) := \dfrac{n_c(t,x)}{N}$, (2.1) can be written as follow:

$$d\varphi(t,x) = (1 - \varphi(t,x))\, J(u)dt \tag{2.2}$$

Now, suppose that the crystal growth in each control volume is independent of other control volumes and is controlled by the identical deterministic model (2.2). However, the crystal growth fluctuations in each control volume and physical phenomena not captured by (2.2). To tackle the previously mentioned problem, we propose to add a stochastic component as part of the nucleation probability (deterministic nucleation rate J) in the crystallized fraction of droplets φ. More precisly, we use $J(u)dt + \sigma(u)dW_t$ isntead of $J(u)dt$. The idea is inspired from Black-Scholes model with a specific diffusion term $\sigma(u)$ to describe the speed of crystallization in Ω.

To motivate our model, we argue as follows. Suppose that the number of crystallized droplets between a control volume and another control in consecutive time intervals $[t, t+\tau)$, $[t+\tau, t+2\tau)$, $\cdots$, $[t+(m-1)\tau, t+m\tau)$ are independent, identically distributed random variables where m is very large. Then by using the Central Limit Theorem the total number of crystallized droplets made in $[t, t+m\tau)$ has approximately a normal law with mean $m\mu$ and variance $m\sigma^2$, where μ and σ^2 indicate the mean and variance of the underlying distribution in time intervals of length $\tau = T/m$. Thus it is reasonable to suppose that the total number of crystallized droplets has a normal law whose mean $J(u)dt$ and variance $\sigma^2(u)dt$.

Therefore we replace $J(u)dt$ in (2.2) by $J(u)dt + \sigma(u)dW_t$, the model becomes stochastic differential equations in φ, and it is to seek the temperature of the emulsion $u(t,\mathbf{x})$ and the ratio of crystallized droplets $\varphi(t,\mathbf{x})$ in the neigh-

bourhood of a point x of the domain Ω and at time $t \in (0,T)$ such that:

$$\begin{cases} \rho c_p \partial_t u - \operatorname{div}(K'(u)\nabla u) &= \rho_0 \mathbb{P} L_F \partial_t \varphi & \text{in } (0,T]\times\Omega, \\ \partial_t \varphi &= (1-\varphi)J(u) + (1-\varphi)\sigma(u)\partial_t W_t, & \text{in } (0,T]\times\Omega, \\ u(0,\cdot) &= u_0 & \text{in } \Omega \\ \varphi(0,\cdot) &= \varphi_0, & \text{in } \Omega, \\ & 0 \leqslant \varphi_0 \leqslant 1, & \text{in } (0,T]\times\Omega \end{cases} \tag{2.3}$$

with $K'(u)$ the thermal conductivity of the emulsion and c_p its specific heat capacity, ρ the mass density, ρ_0 the mass density of water, $\mathbb{P}$ the mass fraction of water and L_F the latent heat of fusion, $\varphi_0 \in L^2(\Omega)$ and $\sigma(u)$ is the corrected of growth crystallization (i.e. diffusion term) and its expression will be specified in Section 6.

The problem model (2.3) is completed with the Fourier boundary conditions on $\Gamma = \partial\Omega$:

$$K'(u)\partial_n u = h_{ext}\left(u - u^{\Gamma}\right), \tag{2.4}$$

where $u^{\Gamma} \in L^{\infty}(\Gamma)$ and h_{ext} are the temperature on the boundary Γ and the thermal exchange coefficient.

Remark 2.1. *The conditions $u_0 := u(0,\cdot) \in L^{\infty}(\Omega)$, $\varphi_0 := \varphi(0,\cdot) \in L^{\infty}(\Omega)$ and $u^{\Gamma} \in L^{\infty}(\partial\Omega)$ relating to the time/boundary will be specified in numerical results if the boundary conditions are changed with respect to the time/boundary of the domain Ω.*

The *a priori* error analysis which we present relies on a crucial technical assumption on the functions defined in (2.3).

2.2. Assumptions

In order to show that the system of equations (2.3) and (2.4) are stable and admit the unique solution, we assume that the following hypotheses concerning data are fulfilled.

(H_1): K a Lipschitz continuous, increasing function satisfies $K(0) = 0$ and its derivatives function is bounded such that $0 < m \leqslant K' \leqslant M$.

(H_2): J is a Lipschitz continuous, positive function satisfying the following inequalities:

$$|J(x) - J(y)| \leqslant \alpha|x-y|, \quad \forall\,(x,y) \in \mathbb{R}^2$$

and

$$J(x) = 0, \quad \forall x \in \mathbb{R}^- \cup [T_m, +\infty[,$$

where $0 < \alpha < \infty$ is a Lipschitz constant.

(H_3): σ is a Lipschitz continuous, positive function satisfying:

$$|\sigma(x) - \sigma(y)| \leqslant \beta|x-y|, \quad (x,y) \in \mathbb{R}^2$$

and

$$\sigma(x) \neq 0, \quad \forall x \in \mathbb{R},$$

where $0 < \beta < \infty$ is a Lipschitz constant.

(H_4): φ_0 is $\mathcal{F}_0$-measurable and $\mathbb{E}\left(\|\varphi_0\|_{0,\Omega}^2\right) < +\infty$.

Remark 2.2. *Assumptions (H_2) and (H_3) are relatively mild, as it is usually fulfilled in the applications we have in mind. Concrete examples will be presented at the end of this paper, as well as in Section 6.*

In the sequel and in order to simplify the study, we assume that $\rho c_p = 1$ and $\rho_0 P L_F = 1$. One gets

Definition 2.3. *A solution of the system of equations* (2.3) *and* (2.4) *is a couple of function (u, φ) in $L^2(\Omega) \times L^2((0,T); L^2(\Theta, H^1(\Omega)))$, such that for all $(t, v) \in [0,T] \times H^1(\Omega)$:*

$$\begin{cases} \int_\Omega (u_t - u_0)\, v\, d\mathbf{x} = \int_0^t \int_\Omega \nabla K(u) \nabla v\, d\mathbf{x} ds + h_{ext} \int_0^t \int_\Gamma \left(u_t - u^\Gamma\right) v\, d\mathbf{x} ds \\ \qquad\qquad\qquad\qquad + \int_\Omega (\varphi_t - \varphi_0)\, v\, d\mathbf{x} \\ u(0,\cdot) = u_0 \quad in \quad \Omega \end{cases} \tag{2.5}$$

and

$$\begin{cases} \varphi_t &= \varphi_0 + \int_0^t (1-\varphi_s) J(u)\, ds + \int_0^t (1-\varphi_s)\sigma(u)\, dW_s, \\ \varphi(0,\cdot) &= \varphi_0 \quad in \quad \Omega. \end{cases} \tag{2.6}$$

In this study, we will adopt the same approach as Vallet and Trujillo in [29] which in turn was inspired by the analysis of existence and uniqueness [1] for a stochastic partial differential equation. More precisely, our strategy is based on *a priori* estimates of solution in L^2-norm and H^1-norm with application of Itô formula and Gronwall's lemma. Next, we exploit the energy estimates for stochastic and parabolic equations in order to prove our main result, see Theorem 4.1.

3. Controlling the *a priori* error of disctretized method

In this section, we will be present some technical lemmas and a discretized method for a stochastic and parabolic partial differential equations.

3.1. Preliminaries and technical lemmas

Here, we present a few isolated technical lemmas that will be used to prove our main result in Section 5. Due to space constraints, these lemmas are presented without proof. For complete proofs of these lemma and some stochastic calculus, we refer to [22] and the references therein. We encourage the reader to see directly the Subsection 3.3, and refer back to technical lemmas from the context of our main result (see Theorem 4.1), where their goal will be evident. We begin this subsection with the statement of Itô formula for process φ_t.

Lemma 3.1. *(Itô formula for process φ_t) Consider real-valued function $f(t,x)$ with second order partial derivatives are continuous. Consider an Itô process $(\varphi_t)_{t\geqslant 0}$ as given by (2.6). Then, with $\sigma(t,\varphi_t) = (1-\varphi_t)\sigma(u)$, one gets*

$$f(t,\varphi_t) - f(0,\varphi_0) = \int_0^t \left(\partial_t f(s,\varphi_s)ds + \frac{1}{2}\sigma^2(s,\varphi_s)\partial_{xx} f(s,\varphi_s)\right) ds + \int_0^t \partial_x f(s,\varphi_s)\sigma(s,\varphi_s)dW_s. \tag{3.1}$$

This is written in shorthand differential form:

$$df(t,\varphi_t) = \left(\partial_t f(t,\varphi_t) + \frac{1}{2}\sigma^2(t,\varphi_t)\partial_{xx} f(t,\varphi_t)\right) dt + \partial_x f(t,\varphi_t)\sigma(t,\varphi_t)dW_t. \tag{3.2}$$

We now cite the discrete Gronwall's (see [42, 43] and the references therein) in order to derive *a priori* estimates for discretized method of stochastic and parabolic differential equations, which help to prove our main results, namely existence and uniqueness of solution (see Theorem 4.1).

Lemma 3.2. *(Discrete Gronwall's) Let μ be a non-negative constant and if (x_n) and (y_n) are non-negative sequences satisfying*

$$x_n \leqslant \mu + \sum_{i=0}^{n-1} x_i y_i, \quad \textit{for all } n \in \mathbb{N}^*.$$

Then, the following inequality holds

$$x_n \leqslant \mu \prod_{i=0}^{n-1} (1+x_i) \leqslant \mu \exp\left(\sum_{i=0}^{n-1} x_i\right), \quad \textit{for all } n \in \mathbb{N}^*. \tag{3.3}$$

In the sequel, we will be used $J(u_n)\delta(1-\varphi_n)$ instead of $J(u_n)(1-\varphi_n)$ to prove and to pass to the limit in *a priori* estimates. Therefore, we prove easy that the obtained solution φ satisfies $0 \leqslant \varphi \leqslant 1$ when N tends to infinity (maximum principle for φ_n).

3.2. Disctretized method

In this section, we will be interest in the discretization of system and the estimating of *a priori* error in order to prove and announce our main result of existence and uniqueness. To do this, we first discretize our system, then we presented some lemmas which allows us to estimate the *a priori* error.

Let $N \in \mathbb{N}^*$. We note $\tau = \frac{T}{N}$, $t_n = n\tau$, $W_n = W_{t_n}$, $u(t_n,x) = u_n$ and $u_n^\Gamma = u^\Gamma(t_n)$ for $n \in \{0,1,2,\cdots,N\}$.

Our starting point is the parabolic equation (2.3), which reads under the simplifying hypothesis on physical properties. Next, we interest in the approximation of the stochastic differential equation, SDE for short (2.3). Here, we apply to SDE an analogue of Euler's method to generate solutions of ordinary

differential equations; it is called Euler-Maruyama method. The discretization has the following way: Find $u_{n+1} \in V$ and $\varphi_{n+1} \in H^1(\Omega)$ $(n = 0, 1, \cdots, N)$ such that for any $v \in V$:

$$\begin{aligned}\int_\Omega (u_{n+1} - u_n)\, v\, dx = \int_0^t \int_\Omega \nabla K(u_{n+1}) \nabla v\, d\mathbf{x} ds + \int_\Omega (\varphi_{n+1} - \varphi_n)\, v\, d\mathbf{x} \\ + h_{ext} \int_0^t \int_\Gamma \left(u_{n+1} - u_n^\Gamma\right) v\, dl ds\end{aligned} \tag{3.4}$$

and

$$\varphi_{n+1} - \varphi_n = (1 - \varphi_n) J(u_n)\tau + (1 - \varphi_n)\sigma(u_n)\left(W_{n+1} - W_n\right). \tag{3.5}$$

with the initial conditions u_0 and φ_0.

3.3. Error analysis and a priori estimates

Lemma 3.3. *For each $n \in \{0, 1, 2, \cdots, N\}$ and for all $\tau > 0$, the equation's systems* (3.4) *and* (3.5) *have with probability one a unique $\mathcal{F}_{n\,\tau}$-measurable $V \times H^1(\Omega)$-valued solutions (u_n, φ_n) in $L^2(\Omega) \times H^1(\Omega)$.*

Proof. The proof based on a fixed point trick, we refer to [29]. The result follows from a classical fixed point argument (see [2, Section 3.11, for example, in the context of stochastic calculus). □

In what follows, we present the lemmas contain some *a priori* estimates, that will be used to study a stochastic/parabolic equations and to show the convergence properties according to our hypothesis. The proof of these lemmas use the some strategy developed in [29].

Remark 3.4. *Error estimates measured in the norm of L^2 and H^1 as in lemma 3.5, 3.6, 3.7 and 3.8 are quite evident and hold very often, see Section 3 and the references therein, in particular, [29].*

Lemma 3.5. *Assume that the hypotheses (H_1), (H_2), (H_3) and (H_4) are hold. Then, for all $\tau > 0$, the following estimates*

$$\sum_{i=1}^{N-1} \mathbb{E}\left(\|\varphi_{i+1} - \varphi_i\|_{0,\Omega}^2\right) \leqslant C, \tag{3.6}$$

$$\sum_{i=1}^{N-1} \mathbb{E}\left(\|\varphi_i\|_{0,\Omega}^2\right) \leqslant \frac{C}{\tau} \tag{3.7}$$

$$\mathbb{E}\left(\|\varphi_n\|_{0,\Omega}^2\right) \leqslant C, \quad n \in \{0, 1, \cdots, N\} \tag{3.8}$$

hold, with a constant $C > 0$ independent of τ and N.

Proof. The estimates of inequalities (3.6), (3.7) and (3.8) are completly similar to those in [29] and they are based on the Lipschitz condition of σ and the Gronwall's Lemma. By multiplying (3.5) by $(\varphi_{n+1} - \varphi_n)$ and by integrating over Ω, the following inequality holds:

$$\mathbb{E}\left(\|\varphi_{n+1}\|_{0,\Omega}^2\right) + \mathbb{E}\left(\|\varphi_{n+1} - \varphi_n\|_{0,\Omega}^2\right) - \mathbb{E}\left(\|\varphi_n\|_{0,\Omega}^2\right) \leqslant D \tag{3.9}$$

with

$$D := \mathbb{E}\left(\int_\Omega (\varphi_{n+1} - \varphi_n)\,(1 - \varphi_n)\sigma(u_n)\,(W_{n+1} - W_n)\;dx\right)$$

We now interest to bound the term D arising the Lipschitz function σ. Since σ is bounded and $\varphi_n\sigma(u_n)$ is $\mathcal{F}_{n\,\tau}$-measurable, and by putting together a maximum principle and Cauchy inequality, the term D is estimated as follows:

$$D \leqslant C_1\tau\mathbb{E}\left(\|\varphi_{n+1}\|_{0,\Omega}^2\right) + C_2\tau \tag{3.10}$$

By using the discrete Gronwall's lemma 3.2, we obtain the desired results.

□

Lemma 3.6. *Assume that the hypotheses (H_1), (H_2), (H_3) and (H_4) are fulfilled. Then, for all $\tau > 0$, the estimates*

$$\mathbb{E}\left(\|\varphi_{n+1} - \varphi_n\|_{0,\Omega}^2\right) \leqslant C\tau, \quad \forall n \in \{0, 1, 2, \cdots, N\} \tag{3.11}$$

holds, with a constant $C > 0$ independent of τ and N.

Proof. Estimate (3.11) is easy to show and is similar that Lemma 3 in [29]. But we use that σ is bounded and Lipschitz function, see hypothesis (H_3). By using (3.5) and a maximum principle, we obtain

$$\begin{aligned}\|\varphi_{n+1} - \varphi_n\|_{0,\Omega}^2 \leqslant\; & 2\tau^2\|J(u_n)\delta\,(1 - \varphi_n)\,\|_{0,\Omega}^2 \\ & + 2\|\sigma(u_n)\,(1 - \varphi_n)\,(W_{n+1} - W_n)\,\|_{0,\Omega}.\end{aligned} \tag{3.12}$$

By computing the expectation of (3.12), we arrive at

$$\mathbb{E}\left(\|\varphi_{n+1} - \varphi_n\|_{0,\Omega}^2\right) \leqslant C\left(\tau + \tau^2\right). \tag{3.13}$$

Since $\tau^2 \leqslant \tau$, the inequality estimate (3.11) holds.

□

In the next lemma, we show the some above inequalities for the sequence $(u_n)_n$, $0 \leqslant n \leqslant N$, that will be used to prove the convergence properties when passing to limit.

Lemma 3.7. *Assume that the hypotheses (H_1), (H_2), (H_3) and (H_4) are fulfilled. Then, for all $\tau > 0$, the following inequalities hold:*

$$\sum_{n=1}^{N-1} \mathbb{E}\left(\|u_{n+1} - u_n\|_{0,\Omega}^2\right) \leqslant C, \tag{3.14}$$

$$\mathbb{E}\left(\|u_n\|_{0,\Omega}^2\right) \leqslant C, \quad \forall\, n = 0, 1, \cdots, N, \tag{3.15}$$

$$\sum_{n=1}^{N-1} \mathbb{E}\left(\|u_n\|_{0,\Omega}^2\right) \leqslant \frac{C}{\tau}, \tag{3.16}$$

$$\sum_{n=1}^{N-1} \mathbb{E}\left(\|u_n\|_{1,\Omega}^2\right) \leqslant \frac{C}{\tau}, \tag{3.17}$$

where $C > 0$ is a constant independent of τ.

Proof. We just have to repeat the proof of lemma in with the special choice for and change two things. In order to prove the inequality estimates, we take $v = u_{n+1}$ as a test function in (3.4), and we use that u_n is $\mathcal{F}_{t_n}$-measurable. Then, the following estimate holds:

$$\frac{1}{2}\mathbb{E}\left(\|u_{n+1}\|_{0,\Omega}^2 - \|u_n\|_{0,\Omega}^2\right) + \frac{1}{4}\mathbb{E}(\|u_{n+1} - u_n\|_{0,\Omega}^2) + m\tau\mathbb{E}(\|u_{n+1}\|_{1,\Omega}^2) \leqslant I + II + C\,\tau$$

where

$$I := \frac{h_{ext}\tau}{2}\mathbb{E}(\|u_{n+1}^{\Gamma}\|_{0,\Gamma}^2), \quad II := \mathbb{E}\left(\|\varphi_{n+1} - \varphi_n\|_{0,\Omega}^2\right).$$

We now sum from $n = 0$ to N, to obtain

$$\frac{1}{2}\mathbb{E}(\|u_{n+1}\|_{0,\Omega}^2) + \frac{1}{2}\sum_{n=0}^{N}\mathbb{E}(\|u_{n+1} - u_n\|_{0,\Omega}^2) + m\tau\sum_{n=0}^{N}\mathbb{E}(\|u_{n+1}\|_{1,\Omega}^2) \leqslant I_{sum} + II_{sum} + C.$$

with

$$I_{sum} = \frac{h_{ext}\tau}{2}\sum_{n=0}^{N}\mathbb{E}(\|u_{n+1}^{\infty}\|_{0,\Gamma}^2) \quad \text{and} \quad II_{sum} = \sum_{n=0}^{N}\mathbb{E}(\|\varphi_{n+1} - \varphi_n\|_{0,\Omega}^2).$$

By using the estimates in Lemma 3.5, we bound the term II_{sum} and we show the announced estimates.

□

In order to announce the main result under our hypotheses (H_1), (H_2), (H_3) and (H_4), we need to bound the H^1-norm and L^2-norm of K arising in the divergence operator. As a consequence of Lemma 3.7 and hypothesis (H_1), we have the following result.

Lemma 3.8. *Assume that* (H_1), (H_2), (H_3) *and* (H_4) *are fulfilled. Then, the following estimates are hold.*

$$\sum_{i=1}^{N-1} \mathbb{E}\left(\|K(u_i)\|_{1,\Omega}^2\right) \leqslant \frac{C}{\tau}, \quad \sum_{i=1}^{N-1} \mathbb{E}\left(\|K(u_{i+1}) - K(u_i)\|_{0,\Omega}^2\right) \leqslant C \tag{3.18}$$

where $C > 0$ *is a constant independent of* τ *and* N.

Proof. Estimate (3.18) is very easy to proof by putting together the Lipschitz condition (H_1) of K and Lemma 3.7. Then, we obtain the desired result. □

4. Existence and uniqueness: second main result

In order to announce our main result of this paper, we consider the following step process:

$$\begin{cases} x_N(t) & = \sum\limits_{i=1}^{N} x_i I_{]t_{i-1},\, t_i]}(t), \quad \forall\, 0 < t \leqslant T, \\ x_N(0) & = x_0. \end{cases} \tag{4.1}$$

where $I_{]t_{i-1},\, t_i]}$ is an indicator function in the interval $]t_{i-1},\, t_i]$ with time step $t_i = i\tau$ (see paragraph 3.2). We are now ready to announce our main results in the following theorem.

Theorem 4.1. *Assume that the hypotheses* (H_1), (H_2), (H_3) *and* (H_4) *are fulfilled. Then, there exists a unique couple of solutions (with probability* 1*)* (u, φ) *in* $L^2(\Omega) \times H^1(\Omega)$ *to problem model* (2.3) *such that:*

(i) φ *has in* H*-continuous trajectories satisfies* $0 \leqslant \varphi \leqslant 1$ *for all* $t \in (0, T)$ *and for all* $x \in \Omega$.

(ii) u *is a continuous function with respect to the initial conditions for all* $(t, x) \in (0, T) \times \Omega$.

Remark 4.2. *For the sake of simplicity, we show Theorem 4.1 in the general case. But, the same result holds in one and two dimensions.*

Theorem 4.1 will be proved in the next section.

5. Proof of the second main result

Our main objective in this section is to prove Theorem 4.1. To do this, we will be show the discrete energy inequalities associated to problem model (2.3). In this case, we use a similar approach described in [29] to show the energy estimates. First, we will start with the discrete energy inequality for stochastic equation. Next, we show the discrete energy inequality for parabolic equation.

5.1. Energy inequality for stochastic equation

The main goal of this subsection is the proof of the discrete energy inequality associated to problem model (2.3). First, let us start with the energy equality in the continuous case for stochastic and parabolic equations. Next, we show the discrete energy inequality for both equations. All are based on Itô's formula, passing to limit and semicontinuity of functional.

In the sequel and for a sake of simplicity, let $K(u_N) = K_N$, $J_N = J(u_N)\delta(1-u_N)$ and $\sigma_N = \sigma(u_N)(1-u_N)$. By applying the notation in (4.1) to (φ_N, u_N) and putting together the previous lemmas of *a priori* estimates, we are ready to announce the following lemma.

Lemma 5.1. *Assume that the hypotheses (H_1), (H_2), (H_3) and (H_4) are fulfilled. Then, the following statements hold.*

(i) A sequence $(\varphi_N)_{N\geqslant 0}$ is bounded in the space $L^2(0,T;L^2(\Theta\times L^2(\Omega))$ and in $L^\infty(\Theta\times H^1(\Omega))$.
(ii) A sequence $(u_N)_{N\in\mathbb{N}}$ is bounded in $L^2(\Theta\times H^1(\Omega))$, $L^2(\Theta;(0,T)\times\partial\Omega)$, $L^\infty(0,T;L^2(\Theta,\partial\Omega))$ and in $L^2(0,T;L^2(\Theta\times H^1(\Omega))$.
(iii) A sequence $(K(u_N))_{N\geqslant 0}$ is bounded in the space $L^\infty(L^2(\Theta\times H^1(\Omega)))$ and in $L^2(0,T;L^2(\Theta\times H^1(\Omega))$.
(iv) A sequence $(J_N)_{N\geqslant 0}$ is bounded in the space $L^2(0,T;L^2(\Theta;H^1(\Omega))$.
(v) A sequence $(\sigma_N)_{N\geqslant 0}$ is bounded in $L^2(0,T;L^2(\Theta,H^1(\Omega)))$.

From the statement of Lemma 5.1, we can therefore extract a subsequence $(u_{N'})_{N'\geqslant 0}$ of $(u_N)_{N\geqslant 0}$ and a subsequence $(\varphi_{N'})_{N'\geqslant 0}$ of $(\varphi_N)_{N\geqslant 0}$ such that the following lemma holds.

Lemma 5.2. *Assume that the hypotheses (H_1), (H_2), (H_3) and (H_4) are fulfilled. There exist $\varphi\in L^2(0,T;L^2(\Theta\times L^2(\Omega))$, u, $\mathcal{K}\in L^2(0,T;L^2(\Theta\times H^1(\Omega))$, $\mathcal{S}\in L^2(0,T;L^2(\Theta\times L^2(\Omega))$ and $\mathcal{J}\in L^2(0,T;L^2(\Theta\times L^2(\Omega))$ such that:*

(i) $\varphi_{N'}\rightharpoonup\varphi$ in $L^2(0,T;L^2(\Theta\times L^2(\Omega))$.
(ii) $u_{N'}\rightharpoonup u$ in $L^2(0,T;L^2(\Theta\times H^1(\Omega))$.
(iii) $K(u_{N'})\rightharpoonup\mathcal{K}$ in $L^2(0,T;L^2(\Theta\times H^1(\Omega))$.
(iv) $J(u_{N'})\delta(1-\varphi_{N'})\rightharpoonup\mathcal{J}$ in $L^2(0,T;L^2(\Theta\times L^2(\Omega))$.
(v) $\sigma(u_{N'})\delta(1-\varphi_{N'})\rightharpoonup\mathcal{S}$ in $L^2(0,T;L^2(\Theta\times L^2(\Omega))$.

In order to make a lecture fluent, we will omit the superscript -'- for N' if there no possibility of confusion. At this point, let start to show the energy inequality for a stochastic equation (2.3). We proceed by the following way: we first inject u_N in (3.5) by adding error $\varepsilon(\tau)$ such that $\varepsilon\longrightarrow 0$ in $L^2(0,T;L^2(\Theta\times L^2(\Omega))$ when $\tau\longrightarrow 0$. Next, we tend N to infinity and we apply Itô's lemma 3.1 to $F(t,\varphi)=e^{-2\gamma t}\varphi$, $\gamma>0$, we therefore get the following equalities:

$$\varphi(t)-\varphi_0=\int_0^t\mathcal{J}ds+\int_0^t\mathcal{S}dW_s \tag{5.1}$$

$$\begin{aligned} dF(t,\varphi) &= -2\gamma F(t,\varphi)dt+e^{-2\gamma t}dF(t,\varphi)\\ &= -2\gamma e^{-2\ t}\varphi dt+\mathcal{J}dt+e^{-2\gamma t}\mathcal{S}dW_t \end{aligned} \tag{5.2}$$

By putting together the integration between 0 and t, and the computing expectation of (5.2), we obtain

$$\mathbb{E}\left(e^{-2\gamma t}\varphi(t)\right) = \mathbb{E}\left(\varphi(0)\right) - 2\gamma\int_0^t e^{-2\gamma s}\varphi ds + \int_0^t \mathcal{J} ds + \int_0^t e^{-2\gamma s}\mathcal{S} dW_s \quad (5.3)$$

We now ready to write a discrete form of (5.3). Then, by using (3.5), the following estimates hold:

$$\begin{aligned}
&-\frac{e^{-2\gamma t_{n+1}}}{2}\mathbb{E}\left(\|\varphi_{n+1}\|_{0,\Omega}^2\right) - e^{-2\gamma t_n}\mathbb{E}\left(\|\varphi_n\|_{0,\Omega}^2\right) - 2\gamma e^{-2\gamma t_{n+1}}\mathbb{E}\left(\|\varphi_{n+1}^2 - \varphi_n^2\right)\\
&\leqslant \frac{e^{2\gamma t_{n+1}} - e^{2\gamma t_n}}{2}\mathbb{E}\left(\|\varphi_n\|_{0,\Omega}^2\right) + e^{-2\gamma t_{n+1}}\tau\mathbb{E}\left(\int_\Omega J(u_n)\delta(1-\varphi_n)\varphi_n\, dx\right)\\
&+ e^{-2\gamma t_{n+1}}\tau\int_\Omega J(u_n)\delta(1-\varphi_n)(\varphi_{n+1}-\varphi_n)\, dx\\
&+ \frac{e^{-2\gamma t_{n+1}}}{2}\mathbb{E}\left(\int_\Omega \sigma(u_n)(1-\varphi_n)(W_{n+1}-W_n)\, dx\right) + \frac{e^{-2\gamma t_{n+1}}}{2}\mathbb{E}\|\varphi_{n+1}-\varphi_n\|_{0,\Omega}^2
\end{aligned}$$

and

$$\begin{aligned}
e^{-2\gamma t_{n+1}}\mathbb{E}\left(\|\varphi_{n+1}\|_{0,\Omega}^2\right) - e^{-2\gamma t_n}\mathbb{E}\left(\|\varphi_n\|_{0,\Omega}^2\right) \leqslant &-2\gamma e^{2\gamma t_n}\mathbb{E}\left(\|\varphi_n\|_{0,\Omega}^2\right) + C\tau^2\\
&+ 2e^{-2\gamma t_{n+1}}\tau\int_\Omega J(u_n)\delta(1-\varphi_n)\varphi_n\, dx\\
&+ \frac{e^{-2\gamma t_{n+1}}}{2}\mathbb{E}\left(\int_\Omega \sigma^2(u_n)(1-\varphi_n)^2\, dx\right)
\end{aligned}$$

By summing from 0 to $(N-1)$ and using Itô isometry, we get

$$\begin{aligned}
e^{-2\gamma t_N}\mathbb{E}\left(\|\varphi_N(T)\|_{0,\Omega}^2\right) \leqslant &\mathbb{E}\left(\|\varphi_0\|_{0,\Omega}^2\right) + C\tau + 2C\tau\left(1 - e^{-2N\gamma\tau}\right)\\
&- 2\gamma\int_0^T e^{-2\gamma t}\mathbb{E}\left(\|\varphi_N\|_{0,\Omega}^2\right) dt + 2\int_0^T e^{-2\gamma t}\mathbb{E}\left(\int_\Omega J_N\varphi_N\, dx\right) dt\\
&+ \int_0^T e^{-2\gamma t}\mathbb{E}\left(\|\sigma_N\|_{0,\Omega}^2\right) dt \qquad (5.4)
\end{aligned}$$

By computing the upper limit for each term in (5.4), we obtain

$$\limsup_{N\longrightarrow\infty} e^{-2\gamma T}\mathbb{E}\left(\|\varphi_N(T)\|_{0,\Omega}^2\right) \leqslant \mathbb{U},$$

with

$$\begin{aligned}
\mathbb{U} := &\mathbb{E}\left(\|\varphi_0\|_{0,\Omega}^2\right) - 2\gamma\limsup_{N\longrightarrow\infty}\int_0^T e^{-2\gamma t}\mathbb{E}\left(\|\varphi_N\|_{0,\Omega}^2\right) dtC\tau\\
&+ \limsup_{N\longrightarrow\infty} 2\int_0^T e^{-2\gamma t}\mathbb{E}\left(\int_\Omega J_N\varphi_N\, dx\right) + \limsup_{N\longrightarrow\infty}\int_0^T e^{-2\gamma t}\int_\Omega \mathbb{E}\left(\sigma_N^2\right)\, dxdt.
\end{aligned}$$

We use that the sequence $\int_0^T J_N dt + \int_0^T \sigma_N dW_t \rightharpoonup \int_0^T \mathcal{J} dt + \int_0^T \mathcal{S} dW_t$ in the space $L^2(\Theta \times L^2(\Omega))$ and thanks to (5.1), to arrive at $\varphi_N(T) \rightharpoonup \varphi(T)$ in $L^2(\Theta \times L^2(\Omega))$. Therefore, if we consider $\eta = \limsup e^{-2\gamma T}\mathbb{E}\left(\|\varphi_N(T)\|_{0,\Omega}^2\right) - e^{-2\gamma T}\mathbb{E}\left(\|\varphi(T)\|_{0,\Omega}^2\right)$, we can see that η is posivite and

$$\eta + e^{-2\gamma T}\mathbb{E}\left(\|\varphi(T)\|_{0,\Omega}^2\right) \leqslant \mathbb{U}. \tag{5.5}$$

Since, for any N in $\mathbb{N}$, the sequences $\|\varphi_N\|_{0,T,\Omega}^2$, $\|\varphi - \varphi_N\|_{0,T,\Omega}^2$, $\|\sigma_N\|_{0,T,\Omega}^2$ and $\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Omega J_N u_N \, dxds\right)$ are bounded in $\mathbb{R}$. Then we can extract a subsequence of each sequence, still indexed by N, such that the previous subsequences converge. Consequently, if one sets

$$\begin{aligned}\mathbb{U}_N = &(1-2\gamma)\|\varphi - \varphi_N\|_{\gamma,T,\Omega}^2 + \|\sigma_N - \sigma(u)(1-\varphi)\|_{0,T,\Omega}^2 \\ &+ 2\int_0^T e^{-2\gamma s}\mathbb{E}\left(\int_\Omega \left(J_N - J(u)\delta(1-\varphi)\right)(u_N - u)dx\right) ds.\end{aligned} \tag{5.6}$$

By using the assumptions (H_2), (H_3), we arrive at

$$\mathbb{U}_N \leqslant (1 - 2\gamma + C_{\alpha,\beta,\delta}\|\varphi - \varphi_N\|_{\gamma,T,\Omega}^2 + C_{\alpha,\beta,\delta}\|u - u_N\|_{\gamma,T,\Omega}^2 \tag{5.7}$$

and by passing to the upper limit in (5.6), we get

$$\begin{aligned}\lim_{N\longrightarrow\infty}\mathbb{U}_N =&\mathbb{U} - \mathbb{E}(\|\varphi_0\|_{0,\Omega}^2) + 2\gamma\|\varphi\|_{\gamma,T,\Omega}^2 - \|\mathcal{S}\|_{\gamma,T,\Omega}^2 + \|\mathcal{S} - \sigma(u)(1-\varphi)\|_{\gamma,T,\Omega}^2 \\ &- 2\int_0^T e^{-2\gamma s}\mathbb{E}\left(\int_\Omega (\mathcal{J}\varphi)\, dx\right) ds.\end{aligned} \tag{5.8}$$

By combining (5.8) and (5.4), we arrive at

$$\lim_{N\longrightarrow\infty}\mathbb{U}_N \geqslant \delta + \|S - \sigma_N\|_{\gamma,T,\Omega}^2 + \lim_{N\longrightarrow\infty}\|\varphi - \varphi_N\|_{\gamma,T,\Omega}^2 \tag{5.9}$$

Thanks to (5.5), we are able to deduce the following inequality

$$\lim_{N\longrightarrow\infty}\mathbb{U}_N \geqslant \delta + \|S - \sigma_N\|_{\gamma,T,\Omega}^2 + \|\varphi - \varphi_N\|_{\gamma,T,\Omega}^2. \tag{5.10}$$

Remark 5.3. *In subsection 5.2, the proof is analogous to the one given in the previous subsection and [29, Section 5.3]. Therefore we only recall the proof with with the special term $\sigma(u)(1-\varphi)$ instead of $b(u,\varphi)$.*

5.2. Energy inequality for parabolic equation

Now we are ready to show the energy inequality for parabolic equation (2.3). To do this, we use the following way: for v in $H^1(Q)$, we construct v^N from the sequence $v_n = v(t_n)$ and we consider $\tilde{v}^N$ the corresponding affine continuous function define from $[0,T]$ to V. To prove this inequality, we use the same strategies in [29] and the previous section. According to (5.1) and thanks to

Lemmas 4 and 2, for all $t \in [0,T]$, we write where $\varepsilon(\tau)$ goes to 0 when τ tends to 0.

$$\int_\Omega (u_N(t)\tilde{v}_N(t) - u_0\tilde{v}_N)\,dx - \int_0^t \int_\Omega u_N \partial_t \tilde{v}_N dxds + \int_0^t \int_\Omega \nabla K(u_N)\nabla v_N dxds$$
$$+h_{ext}\int_0^t \int_\Gamma \left(u_N - u_N^\Gamma\right) v_N\,dlds = \int_\Omega (\varphi_N(t)\tilde{v}_N(t) - \varphi_N(0)\tilde{v}_N(0))\,dx$$
$$- \int_0^t \int_\Omega \varphi_N \partial \tilde{v}_N dxds + \varepsilon(\tau),$$

Let us define $H = H^1(\Omega)$ and $H' = \left(H^1(\Omega)\right)'$. So, if we consider $\mathcal{L}$ define in $H^1(\Omega)$ on $\left(H^1(\Omega)\right)'$ by

$$< \mathcal{L}u, v >_{H',H} = \int_\Omega \nabla K \nabla u\,dx + h_{ext}\int_\Gamma uv\,dl, \tag{5.11}$$

we arrive at $(\mathcal{L}u_N)_N$ converges towards $\mathcal{L}_\infty$ weakly in H' such that

$$< \mathcal{L}_\infty, v >_{H',H} \int_\Omega \mathcal{K}\nabla u\,dx + h_{ext}\int_\Gamma uv\,dl \tag{5.12}$$

Let us define F^Γ in H' by $< F^\Gamma, v >= \int_\Gamma u^\Gamma v\,dl$. Then, for any t, we obtain

$$u(t) - u_0 = \int_0^t \mathcal{L}_\infty v\,ds + \int_0^t F^\Gamma\,ds + \varphi(t) - \varphi_0. \tag{5.13}$$

Consequenlty, whereas u may be chosen continuous with respect to t, we get

$$u_N(T) \rightharpoonup u(T) \quad \text{in} \quad L^2(\Theta; L^2(\Omega)). \tag{5.14}$$

Then, by combining Itô's formula and (5.13), we arrive at:

$$\begin{aligned} e^{-\gamma t}\mathbb{E}\left(\|u(t)^2_{0,\Omega}\right) - \mathbb{E}\left(\|u_0\|^2_{0,\Omega}\right) &= -2\gamma\mathbb{E}\left(\|u\|^2_{0,t,\Omega}\right) - 2\mathbb{E}\left(\int_0^t e^{-2\gamma s} < \mathcal{L}_\infty, u >\,ds\right) \\ &+ 2\mathbb{E}\left(\int_0^t e^{-2\gamma s}\int_\Gamma u^\Gamma u\,dlds\right) \\ &+ 2\mathbb{E}\left(\int_0^t e^{-2\gamma s}\int_\Omega \mathcal{J}u\,dxds\right) \\ &+ \mathbb{E}\left(\int_0^t e^{-2\gamma s}\int_\Omega \mathcal{S}^2\,dxds\right) \end{aligned} \tag{5.15}$$

We proceed with the same in the previous demonstration of sequence φ_N (see subsection 5.1) and we use the *a priori* estimate measured in L^2-norm. We

can write the discrete form of (5.15):

$$
\begin{aligned}
e^{-2\gamma T}\mathbb{E}\left(\|u_N(T)\|_{0,\Omega}^2\right) \leqslant \mathbb{E}\left(\|u_0\|_{0,\Omega}^2\right) + C\tau + 2C\tau\left(1-e^{-2\gamma\tau}\right)\sum_{n=0}^{N-1} e^{-2\gamma n\tau} \\
- 2\gamma\|u_n\|_{0,T,\Omega}^2 - 2\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Omega \nabla K(u_N)\nabla u_N\, dxds\right) \\
- 2\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Gamma (u_N - u_N^\Gamma)u_N\, dlds\right) \\
- \mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Omega J_N u_N\, dxds\right) - \mathbb{E}\left(\int_0^t e^{-2\gamma s}\int_\Omega \sigma_N^2\, dxds\right)
\end{aligned}
$$

Therefore, by computing the upper limit, we arrive at

$$
e^{-2\gamma T}\limsup_{N\longrightarrow\infty}\mathbb{E}\left(\|u_N(T)\|_{0,\Omega}^2\right) \leqslant \Upsilon \tag{5.16}
$$

with Υ the upper limit, as N goes to infinity, of the right-hand-side term in the previous inequality.
Since $u_N(T)$ converge weakly to $u(T)$ in $L^2(\Theta; L^2(\Omega))$, if we set

$$
\upsilon = e^{-2\gamma T}\left(\limsup_{N\longrightarrow\infty}\mathbb{E}\left(\|u_N(T)\|_{0,\Omega}^2\right) - \mathbb{E}\left(\|u(T)\|_{0,\Omega}^2\right)\right), \tag{5.17}
$$

Hence υ is positive, and thanks to (5.15), we arrive at

$$
\upsilon + e^{-2\gamma T}\mathbb{E}\left(\|u(T)\|_{0,\Omega}^2\right) \leqslant \Upsilon. \tag{5.18}
$$

Since for any N in $\mathbb{N}$, the sequences $\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Omega \nabla K(u_N)\nabla u_N\, dxds\right)$, $\|u_n\|_{0,T,\Omega}^2$, $\|u_n\|_{0,T,\Gamma}^2$ and $\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Omega J_N u_N\, dxds\right)$ are bounded in $\mathbb{R}$. Then we can extract a subsequence of each sequence, still indexed by N, such that the above subsequences are convergent. Consequently, by setting

$$
\begin{aligned}
\Upsilon_N = -2\gamma\|u-u_N\|_{0,T,\Omega}^2 - 2\int_0^T e^{-2\gamma s} < \mathcal{L}u - \mathcal{L}u_N, u-u_N > ds \\
+ 2\int_0^T e^{-2\gamma s}\mathbb{E}\int_\Omega \left(J_N - J(u)\delta(1-\varphi)\right)(u_N - u)dxds \\
+ \|u-u_N\|_{0,T,\Omega}^2 + \|\sigma_N - \sigma(u)(1-\varphi)\|_{0,T,\Omega}^2,
\end{aligned}
$$

we have

$$
\Upsilon_N \leqslant C_{\alpha,\beta,\delta}\|\varphi-\varphi_N\|_{0,T,\Omega}^2 + (1-2\gamma+C_{\alpha,\beta,\delta})\|u-u_N\|_{0,T,\Omega}^2 \tag{5.19}
$$

and

$$\lim_{N\longrightarrow\infty}\Upsilon_N = \Upsilon - \mathbb{E}\left(\|u_0\|^2_{0,\Omega}\right) - 2\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Gamma u^\Gamma u\,dxds\right)$$
$$+ 2\mathbb{E}\left(\int_0^T e^{-2\gamma s} < \mathcal{L}u_N, u > \,ds\right) + \|\sigma_N - \mathcal{S}\|^2_{0,T,\Omega} - \|\mathcal{S}\|^2_{0,T,\Omega}$$
$$+ 2\gamma\|u\|^2_{0,T,\Omega} + \lim\|u-u_N\|^2_{0,T,\Omega} - 2\mathbb{E}\left(\int_0^T e^{-2\gamma s}\int_\Omega \mathcal{J}u\,dxds\right)$$

Finaly, we obtain

$$\lim_{N\longrightarrow\infty}\Upsilon_N \geqslant \upsilon + \|\sigma(u)(1-\varphi) - \mathcal{S}\|^2_{0,T,\Omega} + \lim\|u-u_N\|^2_{0,T,\Omega}. \tag{5.20}$$

In the following subsection, we will conclude by putting togther the result of the subsections 5.1 and 5.2.

5.3. End of proof

By combining (5.7) and (5.19), we arrive at

$$\mathbb{U}_N + \Upsilon_N \leqslant (1 - 2\gamma + 2C_{\alpha,\beta,\delta}\left(\|u-u_N\|^2_{0,T,\Omega}\right) \tag{5.21}$$

By choosing $\gamma = 2 + C_{\alpha,\beta,\delta}$, and putting together (5.10) and (5.20), the following estimate holds:

$$\zeta+\eta+2\|\sigma(u)(1-\varphi)-\mathcal{S}\|^2_{0,T,\Omega}+\lim\|\varphi-\varphi_N\|^2_{0,T,\Omega}+\lim\|u-u_N\|^2_{0,T,\Omega} \leqslant 0. \tag{5.22}$$

Consequently, $u_N \longrightarrow u$ in $\mathbb{H}$, $\varphi_N \longrightarrow \varphi$ in $\mathbb{H}$, and $\mathcal{S} := \sigma(u)(1-\varphi)$.

Since, the hypothesis $(H_1) - (H_4)$ is fulfilled, $\delta(x) = \max(0, \min(1, x))$ is a lipschitz continuous function, $K(u)$, $\delta(1-\varphi)$, and $J(u)$ are the corresponding limits of the sequences $(K(u^N))_N$, $(\delta(1-\varphi^N))_N$, and $(J(u^N))_N$. Then, there exist (u, φ) in $\mathcal{V} \times \mathcal{H}$ satisfy (2.6) and (2.5).

Now, it remains to be shown that $0 \leqslant \varphi \leqslant 1$ in order to prove that the couple $(u, \varphi) \in L^2(\Omega) \times L^2((0,T); L^2(\Theta, H^1(\Omega)))$ is a system's solution (2.3) and (2.4). To do this, let g a smooth real function of φ satisfies:

$$g(x) = 0, \quad \forall\, 0 \leqslant x \leqslant 1. \tag{5.23}$$

Then, by using Itô's formula, we have

$$g(\varphi(t)) = g(\varphi_0) + \int_0^t \delta(1-\varphi)J(u)\partial_x(\varphi)\,ds + \frac{1}{2}\int_0^t (1-\varphi)^2\sigma^2(u)\partial_{xx}g(\varphi)\,ds.$$

By passing to expectation, we arrive at

$$\mathbb{E}[g(\varphi(t))] = \mathbb{E}[g(\varphi_0)] + \mathbb{E}\left(\int_0^t J(u)\delta(1-\varphi)\partial_x g(\varphi)\,ds\right)$$
$$+ \frac{1}{2}\mathbb{E}\left(\int_0^t (1-\varphi)^2\sigma^2(u)\partial_{xx}g(\varphi)\,ds\right).$$

From (5.23), the expectation of $g(\varphi(t))$ vanishes for all $0 \leqslant t \leqslant T$. Finally, we conclude that $0 \leqslant \varphi(t) \leqslant 1$ for any $0 \leqslant t \leqslant T$, and (u, φ) is solution to problem model (2.3) with condition (2.4).

5.4. Uniqueness of solutions

The proof is completely classical and based on the ideas of the stochastic energy equality, we refer to [29, Section 5.4] and for more general details [1, Chapter 1]. However, let $(u, \varphi) \in V \times L^2(\Omega)$ and $(\hat{u}, \hat{\varphi}) \in V \times L^2(\Omega)$ be two solutions to problem model (2.3) with condition (2.4) satisfy (2.5) and (2.6). Let $\vartheta = u - \hat{u}$ and $\Psi = \varphi - \hat{\varphi}$. Let us define $I := \mathbb{E}\left(\|\vartheta(t)\|_{0,\Omega}^2 + \|\Psi(t)\|_{0,\Omega}^2 + 2\int_0^t \|\vartheta(s)\|_{1,\Omega}^2 ds\right)$. Then for all $t \in [0, T]$, we obtain

$$\begin{aligned} I \leqslant & -2\mathbb{E}\int_0^t \int_\Omega \Psi(s) J(u) \vartheta(s) dx ds - 2\mathbb{E}\int_0^t \int_\Omega \Psi(s) J(u) \Psi(s) dx ds \\ & + 2\mathbb{E}\int_0^t \int_\Omega [J(u) - J(\hat{u})] \hat{\varphi}(s) \vartheta(s) dx ds + 2\mathbb{E}\int_0^t \int_\Omega [J(\hat{u}) - J(u)] \hat{\varphi} \Psi(s) dx ds \\ & + 2\mathbb{E}\int_0^t \int_\Omega [\sigma(u)(1 - \varphi(s)) - \sigma(\hat{u})(1 - \hat{\varphi}(s)]^2 dx ds. \end{aligned} \tag{5.24}$$

Combining this with $0 \leqslant \varphi \leqslant 1$ and $0 \leqslant \tilde{\varphi} \leqslant 1$, we arrive at

$$\mathbb{E}\left(\|\vartheta\|_{0,\Omega}^2 + 2\int_0^t \|\vartheta(s)\|_{1,\Omega}^2 ds + \|\Psi\|_{0,\Omega}^2\right) \leqslant C\mathbb{E}\left(\int_0^t \left(\|\vartheta(s)\|_{0,\Omega}^2 + \|\Psi\|_{0,\Omega}^2\right) ds\right). \tag{5.25}$$

By applying the continuous Gronwall lemma, we get

$$\mathbb{E}\left(\|\vartheta\|_{0,\Omega}^2\right) = 0, \quad \mathbb{E}\left(\|\Psi\|_{0,\Omega}^2\right) = 0. \tag{5.26}$$

From (5.26), we conclude the uniqueness of solution to problem model (2.3) with condition (2.4).

5.5. Continuity of solution

In order to prove the continuity of the solution (φ, u) with respect to the initial conditions (φ_0, u_0), we just have to repeat the proof of the uniqueness of solution, see subsection 5.4.

6. Numerical experiments, simulations and discussions

Firstly, a short introduction to the experimental device used to measure the experimental temperatures is made. Then, a computational method for solving the physical model based on finite volume method are work out for the stochastic and parabolic differential equations.

6.1. Experimental device and physical model in cylinder

The main objective of this subsection is to present the material drawn from the author's experience ([41, Chapter III, Section III.2], see [2, 20, 28] for many improvements and details) to measure the experimental data in order to validate our mathematical model. This experiment, presented in Figure 1, consists of a metallic cylinder with volume about $500\,cm^3$ (the area contains a water emulsion) and two isolated plugs close it. The thermocouples are located uniformly in the cylinder according to the height h and the radius r of the cylinder. A metallic cylinder is immersed in a liquid solution at temperature $u_b := u_b(t)$, which can be heated or cooled at a constant rate λ using a thermostat. We admit, that the height h of cylinder is very larger than its diameter, to consider that only radius r and time t are the variables of the study. In addition, the experimental conditions are controlled with more precision to crystallize only the water droplets, which are distrubuted homogeneously in the emulsion [2, 20]. In addition, the thermophysical properties are supposed constant during the cooling process. However, the thermophysical properties are dependent on the mass fraction $\mathbb{P}$ of the water droplets dispersed in the emulsion [20], and they vary according to the dispersed phase being in solid, liquid or in an intermediate state (liquid-solid).

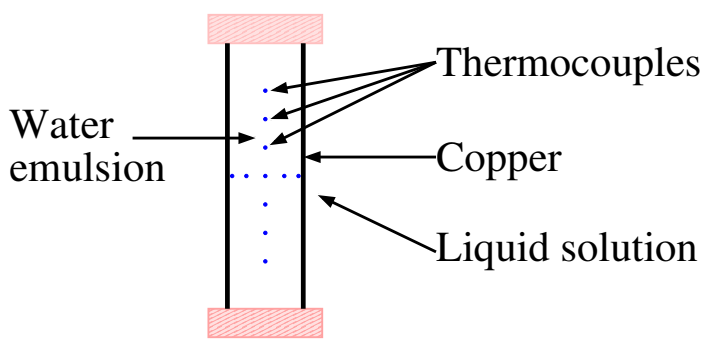

Figure 1: Configuration example of an experimental device.

During the experiment, we have two effects giving the heat transfer inside the emulsion. First one is the cooling process monitored by the so-called bath temperature u^{Γ}. The second one due to the stochastic crystallization of water droplets in emulsion media. Hence, for all $0 < r < R$ (i.e. $\Omega =]0, R[$) and $t > 0$, this process is described by the following equations:

$$\begin{cases} \rho c_p \partial_t u(t,r) & = \frac{1}{r}\partial_r \left(rk\partial_r u(t,r)\right) + \rho_0 \mathbb{P} L_F \partial_t \varphi(t,r), \\ \partial_t \varphi(t,r) & = (1-\varphi(t,r))J(u(t,r)) + \sigma(u(t,r))\partial_t W_t, \end{cases} \tag{6.1}$$

with k the thermal conductivity of the emulsion, ρ the mass density and c_p the specific heat capacity, ρ_0 the mass density of water and , L_F the latent heat of fusion. The functional J and σ are the nucleation probability and diffusion

term which used to modulate the importance of Brownian motion in physical model. Their expressions are given as follows:

$$J(u) = \begin{cases} A\exp\left(-\dfrac{B}{u(u-T_m)^2}\right) & if \quad u < T_m, \\ 0 & otherwise \end{cases} \tag{6.2}$$

and

$$\sigma(u) = \begin{cases} C\exp\left(-\dfrac{D}{u} - \dfrac{E}{(u-T_m)^2}\right) & if \quad u < T_m, \\ 0 & otherwise \end{cases} \tag{6.3}$$

where A, B are assumed to be constant for a given volume of the water droplet [7, 28] and $\Delta u = u - u_m$ is the degree of supercooling. A detailed description of the method to determine $J(u)$ may be found in [7, 15, 28]. For positive constants C and D, the genetic algorithm is adopted to identify the best parameters, in order to fit the experimental temperatures, for more details, see [36].

Remark 6.1. *It is easy to show that the functions J and σ (given by* (6.2) *and* (6.3)*) satisfy the hypotheses* (H_2) *and* (H_3)*, see Subsection 2.2.*

According to the experimental device, the boundary conditions complete the equation (6.1):

$$\partial_r u(t,r)|_{r=0} = 0 \text{ and } -k\partial_r u(t,r)|_{r=R} = h_{ext}\left(u(t,R) - u^{\Gamma}(t)\right) \tag{6.4}$$

where h_{ext} is the convective heat transfer coefficient between the bath and the emulsion. As for initial conditions, we assume: $u(0,r) = u_0$, $\forall r > 0$ and $\partial_n u(t,0) = 0$, $\forall t > 0$. The bath temperature $u_P(t)$ is a linear function with intercept u_0 and :

$$u_p(t) = -\beta t + u_0 \tag{6.5}$$

where $\beta > 0$ is the cooling rate and u_0 is the bath temperature at $t = 0$. During the cooling process, $u_P(t)$ decreases linearly until reaching u_s. Then, $u_P(t)$ is maintained at u_s.

In the next part we will give an alternative method for solving the system of equations (6.1) using the finite volume method (FVM) and considering the initial and boundary conditions.

6.2. Finite volume method for solving physical model

The strategy of finite volume method is to regard the computation domain as subdivided into a collection of finite volumes, see for example [39, 40]. This method involves three basic steps: we first generate the grid by dividing the domain into small discrete control volume and we place the numbers of nodal points in the given domain where each node is surrounded by a control volume or cell, and the physical boundaries coincide with the control volume boundaries. Then, we integrate over the control volume to yield a discretized equation at its nodal point. Finally, we use Tri-diagonal Matrix Algorithm (TDMA), also

as known under the name of Thomas algorithm to solve iteratively the set of discretized equation. But before doing so, we need to introduce an appropriate notations and some considerations.

Let P be a general nodal point of control volume with two neighbours nodes, in one dimension geometry, on the east and west are noted as follow; E and W respectively. We denote by 'e' and 'w', the two faces of the volume control at their sides respectively.

We assume an uniform grid size of the control volumes and we use a full implicit scheme in time with constant time steps Δt.

By integrating the parabolic equation in (6.1) over the span of each control volume at node P, we obtain

$$\begin{aligned}\int_t^{t+\Delta t}\int_V \rho c_p \partial_t u(s,r)\, dV ds = &\int_t^{t+\Delta t}\int_V \frac{1}{r}\partial_r\left(rk\partial_u(s,r)\right)\, dV ds \\ &+ \int_t^{t+\Delta t}\int_V \rho_0 \mathbb{P} L_F \partial_t \varphi\, dV ds.\end{aligned} \tag{6.6}$$

By putting together the SDE of φ, $dV = rdr$ and by integrating between 'w' and 'e', we can get:

$$\begin{aligned}\rho c_p\left(u_P^{t+\Delta t} - u_p^t\right) r_P \Delta r = &\left(rk\frac{\partial u(s,r)}{\partial r}\bigg|_e - rk\frac{\partial u(s,r)}{\partial r}\bigg|_w\right)\Delta t \\ &+ \rho_0 \mathbb{P} L_F\left(1-\varphi_P\right) J(u_P) r_P \Delta r \Delta t \\ &+ \rho_0 \mathbb{P} L_F\left(1-\varphi_P\right)\sigma(u_P)\left(W_{t+\Delta t} - W_t\right) r_P \Delta r.\end{aligned} \tag{6.7}$$

By discretizing the first-order derivative by a second-order central difference scheme, (6.7) can be written as follow:

$$\begin{aligned}\rho c_p\left(u_p^{t+\Delta t} - u_p^t\right) r_P \Delta r = &\left(r_e k_e \frac{u_E^{t+\Delta t} - u_P^{t+\Delta t}}{\delta r_e} - r_w k_w \frac{u_P^{t+\Delta t} - u_W^{t+\Delta t}}{\delta r_w}\right)\Delta t \\ &+ \rho_0 \mathbb{P} L_F\left(1-\varphi_P\right) J(u_P^{t+\Delta t}) r_P \Delta r \Delta t \\ &+ \rho_0 \mathbb{P} L_F\left(1-\varphi_P\right)\sigma(u_P^{t+\Delta t})\left(W_{t+\Delta t} - W_t\right) r_P \Delta r.\end{aligned} \tag{6.8}$$

We can then solve for $u_p^{t+\Delta t}$ by dividing by Δt, and collecting like terms in (6.8):

$$a_P u_P^{t+\Delta t} = a_W u_W^{t+\Delta t} + a_E u_E^{t+\Delta t} + a_P^t u_p^t + T_{deter} + T_{stoch} \tag{6.9}$$

where

$$\begin{aligned}&a_P^t = \rho c_p r_P \frac{\Delta r}{\Delta t}, \quad a_W = k\frac{r_w}{\delta r_w}, \quad a_E = k\frac{r_e}{\delta r_e}, \quad a_p = a_W + a_E + a_P^t, \\ &T_{deter} = \rho_0 P L_F\left(1-\varphi_P\right) J(u_P^{t+\Delta t}) r_P \Delta r, \\ &T_{stoch} = \rho_0 P L_F\left(1-\varphi_P\right)\sigma(u_P^{t+\Delta t}) r_P \frac{\Delta r}{\Delta t}\left(W_{t+\Delta t} - W_t\right),\end{aligned}$$

For the nodes adjacent to borders, they have a face value given by replacing the border conditions (6.4) in (6.9).

We now interest in the stochastic term T_{stoch} containing the Brownian motion. By using $W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$ and $\mathcal{N}(0, \Delta t) \sim \sqrt{\Delta t}\mathcal{N}(0, 1)$, T_{stoch} can be expressed in law as follow:

$$T_{stoch} \sim \rho_0 P L_F \left(1 - \varphi_P\right) \sigma(u_P^{t+\Delta t}) r_P \frac{\Delta r}{\sqrt{\Delta t}} \mathcal{N}(0, 1) \tag{6.10}$$

where $\mathcal{N}(0, 1)$ is a standard normal random with mean 0 and variance 1. Then, by putting $Z \sim \mathcal{N}(0, 1)$, we can write

$$T_{stoch} \sim \rho_0 P L_F \left(1 - \varphi_P\right) \sigma(u_P^{t+\Delta t}) r_P \frac{\Delta r}{\sqrt{\Delta t}} Z \tag{6.11}$$

To simulate a standard normal random Z, we adopt a Box-Muller method [37] which based on uniform distribution, $\mathcal{U}(0, 1)$. Because, it uses the logarithmic function and trigonometric computational, whereas algorithms based on the inverse transformation usually use simple polynomial evaluations. Moreover, its is really simple in term of programming and this is why I think it is a really good one.

The Box-Muller method generates a pair of independent standard normal random variables using a pair of uniform random numbers in two steps:

- U_1 and U_2 be two independent variables uniformly distributed on $(0, 1)$;
- Compute $Z_1 = \sqrt{-\log(U_1)}\cos(2\pi U_2), \quad Z_2 = \sqrt{-\log(U_1)}\sin(2\pi U_2)$.

By concatenating Z_1 and Z_2, we finally simulate the standard random variable Z.

To finish out the finite volume method, we use TDMA method to solve iteratively the above discretized equations.

6.3. Simulations and discussions

In this subsection, we present two numerical tests in order validate our one dimensional mathematical model and its theoretical results, and to ensure that the stochastic model developed in this work have a good behaviour for crystallization within water emulsion. The first test is devoted to deterministic case when σ vanishes and the second one concern the proposed model (6.1).

The thermo-physical properties and test-case are taken from [41]. We take $\mathbb{P} = 0.50$, $T_m = 0\,^\circ C$, $T_s = -68\,^\circ C$, $\beta = -18.8^\circ C/h$ and $R = 27.5\,cm$. The numerical simulations are carried out with the FORTRAN language and developed by our team.

6.3.1. Deterministic case

In this first case, we will choose the diffusion term σ vanishes for any $(t, x) \in [0, T] \times \Omega$. In terms of σ given in (6.3), we choose $C = 0$. Then, we retrieve the numerical temperature profiles for the mathematical model in derteministic

case, see [41, Chapter IV, Subsection IV.4.2]. Figure 2 show the numerical tempretures at different values of r and the batch temperature u_p. We therefore obtain the computed temperature u is very close to the experimental one found in [41]. In what follows, we present the ratio crystallized droplets φ at different

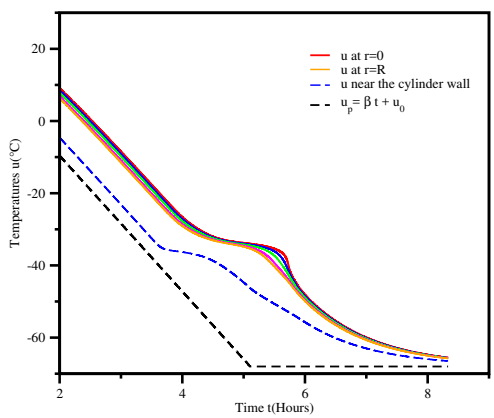

Figure 2: Deterministic case: bath temperature u_p and Temperature profiles $u(t,r)$ at different values of r.

values of r, see Figure 3. We show that the function φ satisfies $0 \leqslant \varphi(t,r) \leqslant 1$, for all $(t,r) \in [0,T] \times]0,R[$.

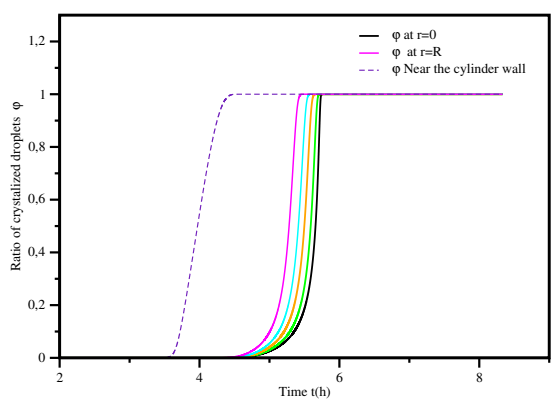

Figure 3: Derterministic case: ratio of crystallized droplets φ at different values of r.

6.3.2. Stochastic case

In this second case, we choose $C = 0.1$, $D = E = 0$. The following figures show that the stochastic character for the temperature u and the ratio crystallized droplets φ at the plateau region, see Figures 4 and 5.

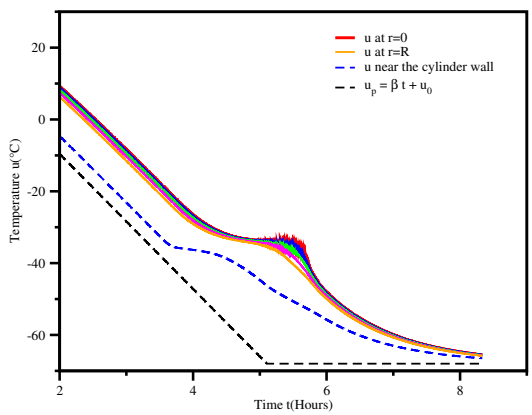

Figure 4: Stochastic case: bath temperature u_p and Temperature profiles $u(t, r)$ at different values of r.

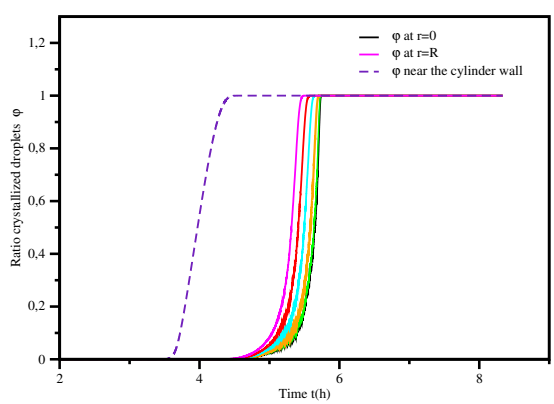

Figure 5: Ratio of crystallized droplets φ at different values of r.

In what follows, we present in Figure 6 the temperature u and in Figure 7 the Ratio of crystallized droplets φ at different values of r with $C = 30$, $D = 2200$ and $E = 5 \times 10^7$.

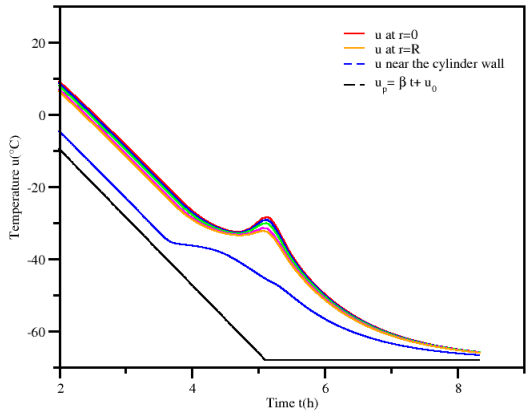

Figure 6: Bath temperature u_p and Temperature profiles $u(t, r)$ at different values of r.

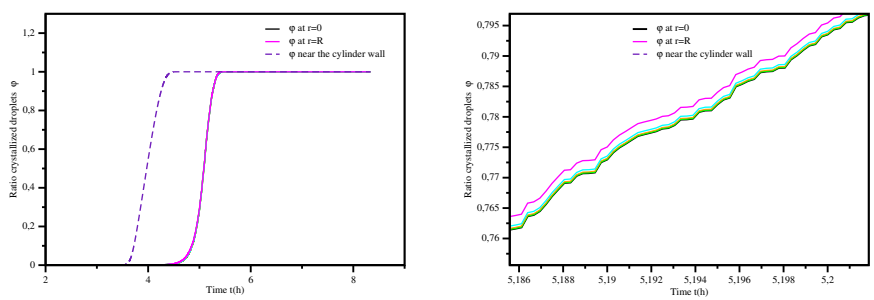

Figure 7: Ratio of crystallized droplets φ (left) and its room (right) at different values of r.

Remark 6.2. *In order to choose the best parameters C, D and D to fit the experimental temperature of Krichi [41], we use the Genetic algorithm described in our paper [55].*

For a sake of simplicity, we choose the parameters $C = 31.7885$, $D = 2.3156 \times 10^3$ and $E = 5.8371 \times 10^7$ defined by Genetic algorithm. Then, we are now ready to show the comparison between the output numerical temperatures u of our mathematical model (6.1) and the experimental temperature given by Krichi [41]. Figure 8 shows the variation of the emulsion temperature at the center ($r = 0$) and extremity ($r = R$) of the cylinder versus time. From a qualitative point of view, we get the computed temperature (solid line) is close to the experimental one (dashed line). In this case, we retrieve numerically the growth crystallization of droplets, see [41, Chapter IV, Subsection IV.3.2]. As intended, the water droplets near the inner face of the cylinder begin to crystallize first

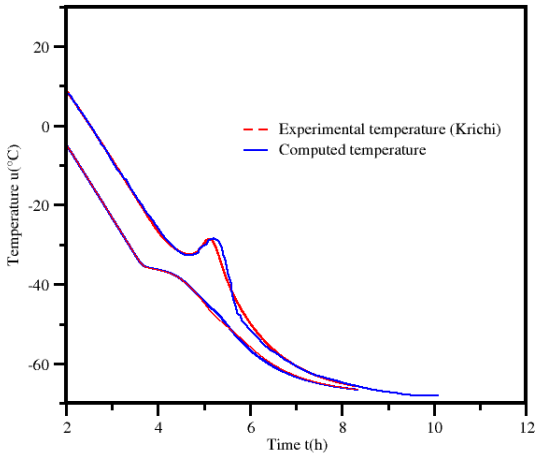

Figure 8: Comparison between experimental and numerical temperatures.

at around the temperature of $-38°C$. The thermal energy released during the solidification of these droplets heats the central region of the cylinder, causing a significant delay in the water droplets crystallization in this zone. One can also remark that the droplet crystallization in the cylinder's center ($r = 0$) has caused an important increase in the emulsion temperature. Such temperature increase can be explained by a rapid crystallization of a large quantity of water droplets, which can cause a chaotic droplet movement in the emulsion, leading to coalescence phenomenon.

7. Conclusion

The main goal of this paper consisted in proposing a close-to-reality stochastic model of crystallization process in dispersed media by means of stochastic differential equations. As the first step, a mathematical modeling based on SDE is done. The second step, we have proved the uniqueness and existence of solution to stochastic differential equations. a Finite Volume method is used to solve the direct physical model. This method is based on a full implicit scheme to discretize a heat transfer equation and a Box-Muller method to simulate a Brownian motion. All simulations results demonstrated that the proposed model is accurate and efficient when compared with the experimental temperature. Future works are intended to be done in this field, such as the use of a second backward stochastic differential equations (2BSDE), in order to describe the crystallization and fit the measured temperature.

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